

Project 5

Smart Financial Services Kiosk with AI Advisory System

Activity Listing

Activity ID	Activity Name	Description	Duration (Days)	Predecessor(s)	Responsible Role
A1	Project Initiation	Define project objectives, scope, and team responsibilities	2	-	Project Manager
A2	Project Charter Preparation	Prepare and approve the project charter	1	A1	Project Manager
A3	Requirement Gathering	Collect functional and non-functional requirements	3	A2	Business Analyst
A4	Feasibility Study	Analyze technical, economic, and operational feasibility	2	A3	Project Manager
A5	System Architecture Design	Design overall system architecture	2	A4	System Designer
A6	Database Design	Design database schema and tables	2	A5	Database Designer
A7	User Interface Design	Design kiosk screens and navigation	2	A5	UI/UX Designer
A8	AI Advisory Module Design	Design AI recommendation logic	2	A5	AI Developer
A9	Development Environment Setup	Install software tools and configure environment	1	A5	Software Developer
A10	User Registration Module	Develop user information module	2	A9	Software Developer
A11	Income & Expense Analysis Module	Develop budgeting and expense analysis functions	3	A10	Software Developer

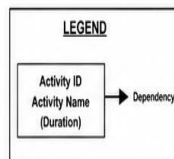
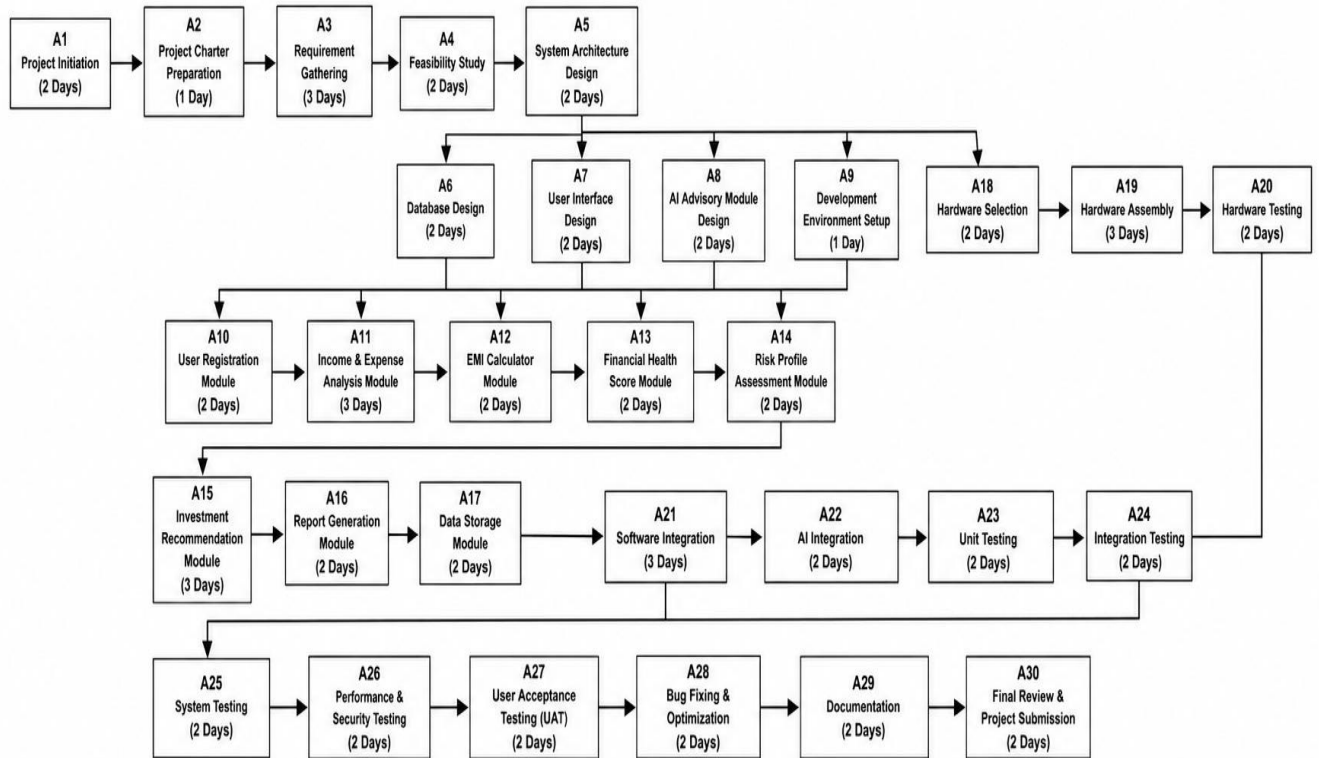
A12	EMI Calculator Module	Develop EMI calculation functionality	2	A11	Software Developer
A13	Financial Health Score Module	Implement financial health scoring logic	2	A12	Software Developer
A14	Risk Profile Assessment Module	Develop risk profile analysis	2	A13	AI Developer
A15	Investment Recommendation Module	Develop AI-based investment recommendations	3	A14	AI Developer
A16	Report Generation Module	Generate financial reports (Excel/PDF)	2	A15	Software Developer
A17	Data Storage Module	Store and retrieve user data securely	2	A16	Database Developer
A18	Hardware Selection	Select Raspberry Pi, display, and peripherals	2	A5	Hardware Engineer
A19	Hardware Assembly	Assemble kiosk hardware components	3	A18	Hardware Engineer
A20	Hardware Testing	Verify hardware functionality	2	A19	Hardware Engineer
A21	Software Integration	Integrate all software modules	3	A17, A20	Software Developer
A22	AI Integration	Integrate AI advisory engine with application	2	A21	AI Developer
A23	Unit Testing	Test individual software modules	2	A22	Testing Engineer
A24	Integration Testing	Verify interaction between modules	2	A23	Testing Engineer
A25	System Testing	Perform complete system testing	2	A24	Testing Engineer
A26	Performance & Security Testing	Test response time, reliability, and data security	2	A25	Testing Engineer

A27	User Acceptance Testing (UAT)	Validate system with end users	2	A26	End User & Project Manager
A28	Bug Fixing & Optimization	Resolve defects identified during testing	2	A27	Development Team
A29	Documentation	Prepare technical documentation and user manual	2	A28	Documentation Engineer
A30	Final Review & Project Submission	Final verification, presentation, and project handover	2	A29	Project Manager

Network Diagram

Smart Financial Services Kiosk with AI Advisory System

Activity-on-Arrow (AOA) Network Diagram



- NOTES**
- Numbers in brackets indicate duration in days.
 - This is an Activity-on-Arrow (AOA) network diagram.
 - Project duration (critical path) = 48 Days.

Critical Path Method (CPM) Calculation

Act. ID	Activity Name	Dur. (Days)	Predecessor(s)	ES	EF	LS	LF	Total Float	Critical
A1	Project Initiation	2	-	0	2	0	2	0	Yes
A2	Project Charter Preparation	1	A1	2	3	2	3	0	Yes
A3	Requirement Gathering	3	A2	3	6	3	6	0	Yes
A4	Feasibility Study	2	A3	6	8	6	8	0	Yes
A5	System Architecture Design	2	A4	8	10	8	10	0	Yes
A6	Database Design	2	A5	10	12	13	15	3	No
A7	User Interface Design	2	A5	10	12	10	12	0	Yes
A8	AI Advisory Module Design	2	A5	10	12	12	14	2	No
A9	Development Environment Setup	1	A5	10	11	12	13	2	No
A10	User Registration Module	2	A9	11	13	13	15	2	No
A11	Income & Expense Analysis Module	3	A10	13	16	15	18	2	No
A12	EMI Calculator Module	2	A11	16	18	18	20	2	No
A13	Financial Health Score Module	2	A12	18	20	20	22	2	No
A14	Risk Profile Assessment Module	2	A13	20	22	22	24	2	No
A15	Investment Recommendation Module	3	A14	22	25	24	27	2	No
A16	Report Generation Module	2	A15	25	27	27	29	2	No
A17	Data Storage Module	2	A16	27	29	29	31	2	No

A18	Hardware Selection	2	A5	10	12	19	21	9	No
A19	Hardware Assembly	3	A18	12	15	21	24	9	No
A20	Hardware Testing	2	A19	15	17	24	26	9	No
A21	Software Integration	3	A17, A20	29	32	31	34	2	No
A22	AI Integration	2	A21	32	34	34	36	2	No
A23	Unit Testing	2	A22	34	36	36	38	2	No
A24	Integration Testing	2	A23	36	38	38	40	2	No
A25	System Testing	2	A24	38	40	40	42	2	No
A26	Performance & Security Testing	2	A25	40	42	42	44	2	No
A27	User Acceptance Testing	2	A26	42	44	44	46	2	No
A28	Bug Fixing & Optimization	2	A27	44	46	46	48	2	No
A29	Documentation	2	A28	46	48	48	50	2	No
A30	Final Review & Project Submission	2	A29	48	50	50	52	2	No

Critical Path

The activities with **zero total float** form the **Critical Path**:

A1 → A2 → A3 → A4 → A5 → A7

These activities must be completed on schedule because any delay will directly affect subsequent dependent activities.

Project Summary

Parameter	Value
Total Activities	30
Project Duration (Planned)	48 Days
Critical Activities	6
Non-Critical Activities	24
Maximum Float	9 Days (Hardware Activities)
Minimum Float	0 Days
Scheduling Technique	Critical Path Method (CPM)